ST ALOYSIUS (DEEMED TO BE UNIVERSITY)

Mangaluru

SCHOOL OF ENGINEERING

(UG Programme)

II Internal Exam – November 2025

BTech (AIML/ECE) - Semester I

PROBLEM SOLVING THROUGH 'C' PROGRAMMING

Time: 1½ hrs. **Max Marks: 50**

Note:

1. This question paper consists of 2 sections. PART- A and PART- B.
2. Answer FOUR questions from PART B either **8 OR 9** and **10 OR 11.**

PART – A

I Answer any FIVE of following questions: **(5 x 2 =10)**

1. Explain how elements are stored and accessed in a two-dimensional array with a suitable example.
2. Write a C program to ~~swap~~ Sum two numbers using pointers.
3. Explain how recursion works in function calls with an example.
4. Describe how strings are stored and terminated in memory in C.
5. Explain how gets() and scanf() functions are used to read strings in C. Differentiate between the two with an example.
6. Differentiate between call by value and call by reference.
7. Demonstrate how a file can be opened in read mode in C with proper syntax and explanation.

PART – B

(4 x 10 =40)

II Answer the following:

8. a. Explain the declaration, initialization, and accessing of one-dimensional and two-dimensional arrays with suitable examples. **(10)**

   b. Explain different methods of passing parameters to functions in C with examples and write a note on Recursive functions. **(10)**

OR

9. a. Explain about the significance of Functions. Elaborate in detail about the different elements of a function with suitable examples. **(1**

b. Write a C program to perform a binary search for a given key integer in a single dimensional array of numbers in ascending order and report success or failure in the form of a suitable message array and the sorted array with suitable headings.

III Answer the following

10. a. Explain various string handling functions in C (strcpy, strlen, strcmp, strcat, strrev) with examples.

b. Explain the concept of pointers with examples. Write a program to swap two numbers using pointers.

OR

11. a. Explain the different file operations in C and write a C program to apply these operations for reading from a file and writing to a file.

b. Describe Sorting in arrays. Write a C program, to implement a bubble sort technique using function to sort given N integers in ascending/ descending order as per user's preference.

Mapping of Questions with Bloom's Level and Course Outcomes (COs)

PART	I							II				III			
Q. No.	1	2	3	4	5	6	7	8a	8b	9a	9b	10a	10b	11a	11b
Blooms Level	2	3	2	2	2	2	2	2	3	2	3	2	3	3	3
Mapped CO(s)	3	4	3	3	3	3	4	3	3	3	3	3	4	4	3

Date:06.11.2025 **Time:10.00-11.30** **Copies**